Unit 2: Random Walks, Stationarity & Unit Root Tests

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Outline

Learning Objectives

By the end of this unit you will be able to:

1. Simulate white noise and random walk processes in R
2. Explain why the random walk is non-stationary
3. State the conditions for weak (covariance) stationarity
4. Apply the ADF and KPSS tests to diagnose unit roots
5. Interpret ACF and PACF plots for model identification

Companion script
Open `Unit-2-RandomWalks-Stationarity.R` — it contains worked examples and Monte Carlo simulations you run alongside these slides.

White Noise

White Noise Process

A **white noise** process $\{\epsilon_t\}$ satisfies:

1. $E[\epsilon_t] = 0$ (zero mean)
2. $\text{Var}(\epsilon_t) = \sigma^2$ (constant variance)
3. $\text{Cov}(\epsilon_t, \epsilon_s) = 0$ for $t \neq s$ (no serial correlation)

White noise is the simplest stationary process and the "building block" for every time series model we encounter today.

In R

```
set.seed(42)
e <- rnorm(100)
```

generates $\epsilon_t \sim N(0, 1)$

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Random Walks

A **random walk** is defined as:

$$y_t = y_{t-1} + \epsilon_t, \qquad \epsilon_t \sim WN(0, \sigma^2)$$

By recursive substitution: $y_t = y_0 + \sum_{i=1}^{t} \epsilon_i$

Key properties:

- $E[y_t] = y_0 \quad$ (constant mean)
- $\text{Var}(y_t) = t \cdot \sigma^2 \quad$ (grows with t)
- Changes $\Delta y_t = \epsilon_t$ are white noise

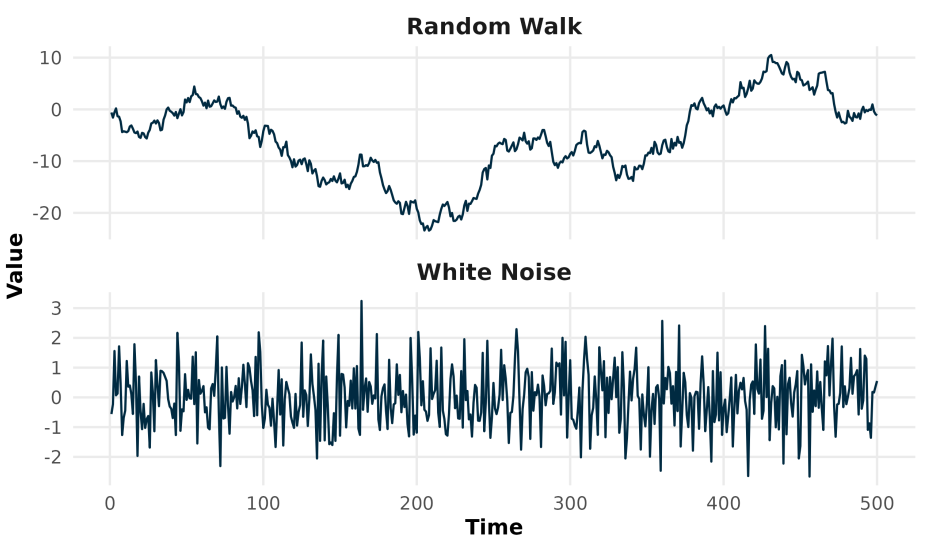
The growing variance makes the process **non-stationary**.

Random walk with drift
Adding a constant μ:

$y_t = \mu + y_{t-1} + \epsilon_t$

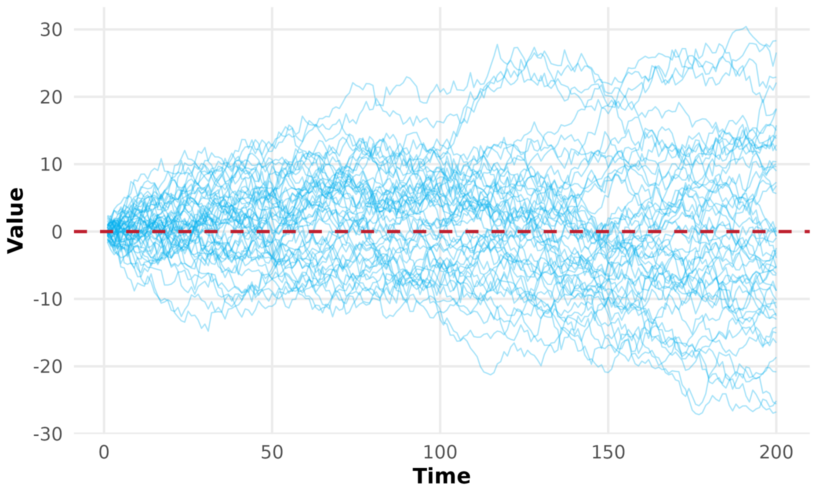
introduces a deterministic trend. Variance still grows $\Rightarrow$ still non-stationary.

White Noise vs Random Walk



White noise fluctuates around zero with constant variance. The random walk (its cumulative sum) wanders

Multiple Realisations



Each realisation looks very different — this is the nature of a non-stationary process. The “fan” of paths widens over time because $\mathrm{Var}(y_t) = t \cdot \sigma^2$.

Simulating Random Walks in R

Single random walk:

```
e <- rnorm(100)
y <- cumsum(e)
plot(y, type = "l")
```

With drift:

```
mu <- 0.5
y <- cumsum(mu + e)
```

Multiple realisations:

```
1:30 %>%
  map(~create_random_walk(.)) %>%
  list_rbind()
```

Financial interpretation
The Efficient Market Hypothesis (weak form) implies stock prices follow a random walk.

The AR(1) Process

General AR(1) Process

The first-order autoregressive process:

$$y_t = \alpha + \beta\, y_{t-1} + \epsilon_t$$

Condition	**Behaviour**	**Stationary?**	**Moments**
$\lvert\beta\rvert < 1$	Mean-reverting	Yes	$E[y] = \frac{\alpha}{1-\beta}$, $\mathrm{Var}(y) = \frac{\sigma^2}{1-\beta^2}$
$\beta = 1$	Random walk	No	Variance $\to \infty$
$\lvert\beta\rvert > 1$	Explosive	No	Diverges

The parameter β determines everything: **stationarity** requires $\lvert\beta\rvert < 1$.

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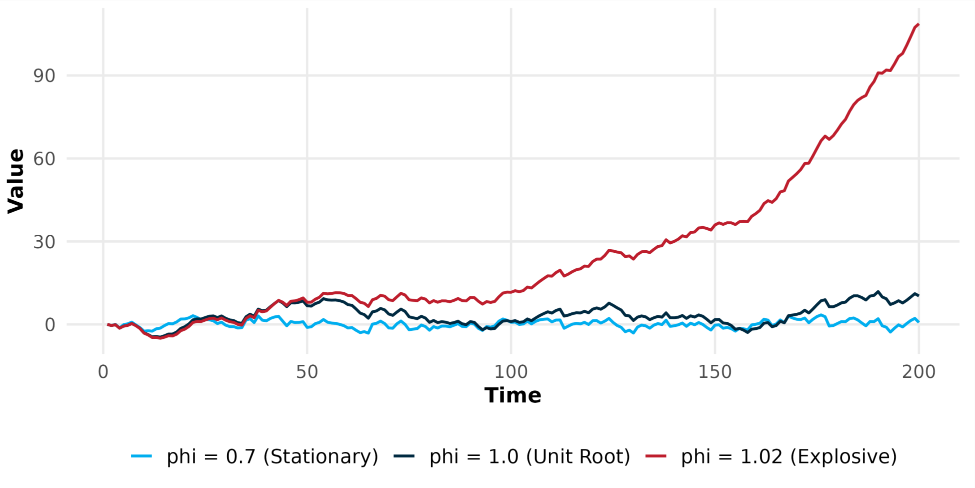
AR(1): Three Regimes



Left: the stationary process reverts to its mean. Centre: the unit-root process wanders. Right: the explosive process diverges rapidly. All three use the *same* white noise innovations.

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Finite Sample Bias

Why estimated $\hat{\beta}$ is biased downward

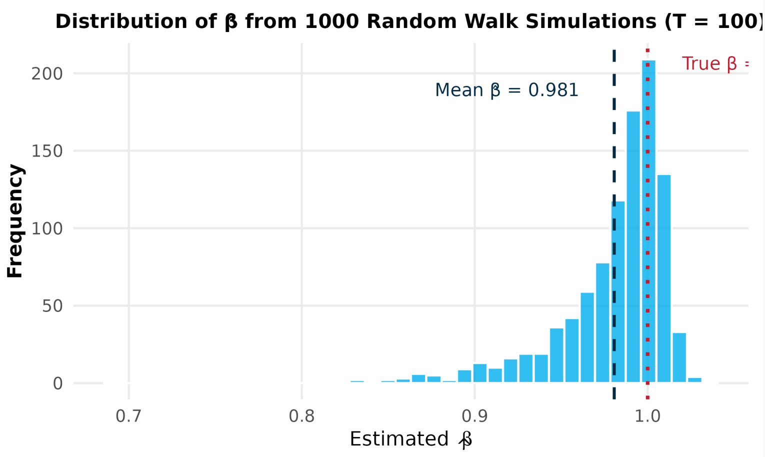
When estimating $y_t = \hat{\beta} y_{t-1} + \hat{\epsilon}_t$ from a random walk:

- True $\beta = 1$, but $\hat{\beta}$ is typically **less than 1**
- This is the **finite sample bias** (also called Nickell bias)
- Bias $\approx -7/T$ for an AR(1) without intercept
- Disappears as $T \to \infty$

Monte Carlo verification

Simulate 1000 random walks, estimate $\hat{\beta}$ for each, and plot the distribution. See the R script, Part D.

Monte Carlo Results



The distribution of $\hat{\beta}$ is centred just below 1 (red dotted line), confirming the downward finite-sample bias. This is why we need **formal unit root tests** rather than simply checking if $\hat{\beta} \approx 1$.

Stationarity

Weak (Covariance) Stationarity

A time series $\{y_t\}$ is **weakly stationary** if:

1. $E[y_t] = \mu$ (constant mean)
2. $\mathrm{Var}(y_t) = \sigma^2 < \infty$ (constant, finite variance)
3. $\mathrm{Cov}(y_t, y_{t-k}) = \gamma_k$ (autocovariance depends only on lag k)

Why does stationarity matter?

- Most time series models (ARMA, GARCH) **require** stationarity
- Non-stationary data $\Rightarrow$ spurious regressions, misleading inference
- Solution: **differencing** ($\Delta y_t = y_t - y_{t-1}$) or **de-trending**

Quick Check: Stationary or Not?

For each process, decide whether it is stationary:

1. $y_t = 0.5 + 0.3\, y_{t-1} + \epsilon_t$
2. $y_t = y_{t-1} + \epsilon_t$
3. $\Delta y_t = \epsilon_t$
4. $y_t = 10 + 2t + \epsilon_t$ (deterministic trend)

Hint: Check whether mean and variance are constant over time. Which conditions from the previous slide are violated?

Answers: (1) Stationary ($|\beta| = 0.3 < 1$). (2) Non-stationary (random walk, Var $\to \infty$). (3) Stationary (the *changes* are white noise). (4) Non-stationary (mean grows with t).

Unit Root Tests

Augmented Dickey–Fuller (ADF) Test

Tests the null hypothesis of a unit root:

$$\Delta y_t = \alpha + \delta t + \gamma y_{t-1} + \sum_{j=1}^{p} \phi_j \Delta y_{t-j} + \epsilon_t$$

H_0	$\gamma = 0$ (unit root)
H_1	$\gamma < 0$ (stationary)
Reject H_0	Evidence of stationarity
Fail to reject	Cannot reject unit root

In R

```
tseries::adf.test(y)
```

The KPSS test reverses the hypotheses:

H_0	Series is stationary
H_1	Series has a unit root

Best practice: Use *both* tests together.

ADF	KPSS	Conclusion
Reject	Fail to reject	Stationary
Fail to reject	Reject	Non-stationary
Both reject		Inconclusive
Both fail		Inconclusive

How Many Differences?

The **`feasts`** package can automatically determine the number of differences needed for stationarity:

```
data %>% features(y, unitroot_ndiffs)
```

- Returns 0 if the series is already stationary
- Returns 1 if one difference is needed
- Returns 2 if two differences are needed (rare)

The "I" in ARIMA

ARIMA(p, d, q): the parameter d is the number of differences applied to achieve stationarity.

$d = 0$: already stationary

$d = 1$: first difference

$d = 2$: second difference

ACF and PACF

Autocorrelation Function (ACF)

The autocorrelation at lag k measures the linear dependence between y_t and y_{t-k}:

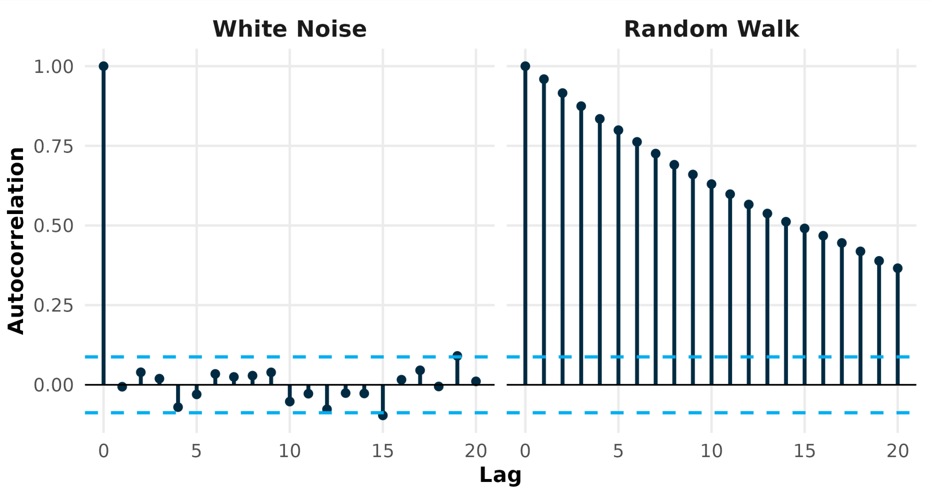
$$r_k = \frac{\sum_{t=k+1}^{T}(y_t - \bar{y})(y_{t-k} - \bar{y})}{\sum_{t=1}^{T}(y_t - \bar{y})^2}$$

- $r_0 = 1$ always
- White noise: $r_k \approx 0$ for $k \geq 1$
- Random walk: r_k decays very slowly

In R

```
data %>%
  gg_tsdisplay(y,
    plot_type = "partial",
    lag_max = 20)
```

ACF: White Noise vs Random Walk



Left: white noise autocorrelations fall within the 95% confidence band (blue dashed lines) — no significant dependence. Right: the random walk ACF decays very slowly, indicating high persistence and non-stationarity.

Partial Autocorrelation Function (PACF)

The PACF at lag k measures the correlation between y_t and y_{t-k} *after removing* the linear effects of lags $1, \ldots, k-1$.

Process	**ACF**	**PACF**
AR(p)	Decays gradually	Spikes at lags 1..p
MA(q)	Spikes at lags 1..q	Decays gradually
ARMA(p,q)	Decays gradually	Decays gradually

Reading the plots
Cuts off: abrupt drop to zero (within confidence bands)

Tails off: slow, gradual decay

We apply this table extensively in Unit 3.

Exercises

Open `Unit-2-RandomWalks-Stationarity.R` — the script walks you through complete examples before each exercise.

1. **AR(1) simulation:** Simulate 200 observations for each case and plot: (a) $\alpha = 2$, $\beta = 1$ (drift); (b) $\alpha = 0$, $\beta = 0.8$ (stationary); (c) $\alpha = 0$, $\beta = 1.05$ (explosive). Classify each as stationary, unit root or explosive.
2. **Theoretical moments:** For the stationary case ($\beta = 0.8$), compute $E[y] = \alpha/(1-\beta)$ and $\text{Var}(y) = \sigma^2/(1-\beta^2)$. Compare with `mean(y)` and `var(y)` from your simulation.
3. **Real data:** (a) Download a stock of your choice using `download_data("stock_prices", symbols = "AAPL", …)`; (b) compute log returns with `mutate()` and `difference()`; (c) run KPSS (`unitroot_kpss`) and ADF (`adf.test`) tests; (d) plot the ACF/PACF — is the return series stationary?

Hints: Starter code for the AR(1) loop is provided in the script. Use `feasts::unitroot_kpss` and `tseries::adf.test`.

Summary

You can now:

- Simulate white noise and random walks and explain why they differ
- Classify an AR(1) process as stationary, unit root or explosive based on β
- State the three conditions for weak stationarity
- Apply the ADF test (H_0: unit root) and KPSS test (H_0: stationary) and interpret them jointly
- Read ACF/PACF plots to distinguish AR and MA patterns

Next: Unit 3 — ARMA/ARIMA Models & Forecasting